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وزارة التعليم العالي و
البحث العلمي
جامعة قرطاج
المدرسة التونسية للتقنيات

Graduation Project Report

To obtain :

The National Engineering Diploma from the Ecole Polytechnique
de Tunisie

Decomposition AI for thermal satellite images resolution enhancement

Elaborated by:

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Within:



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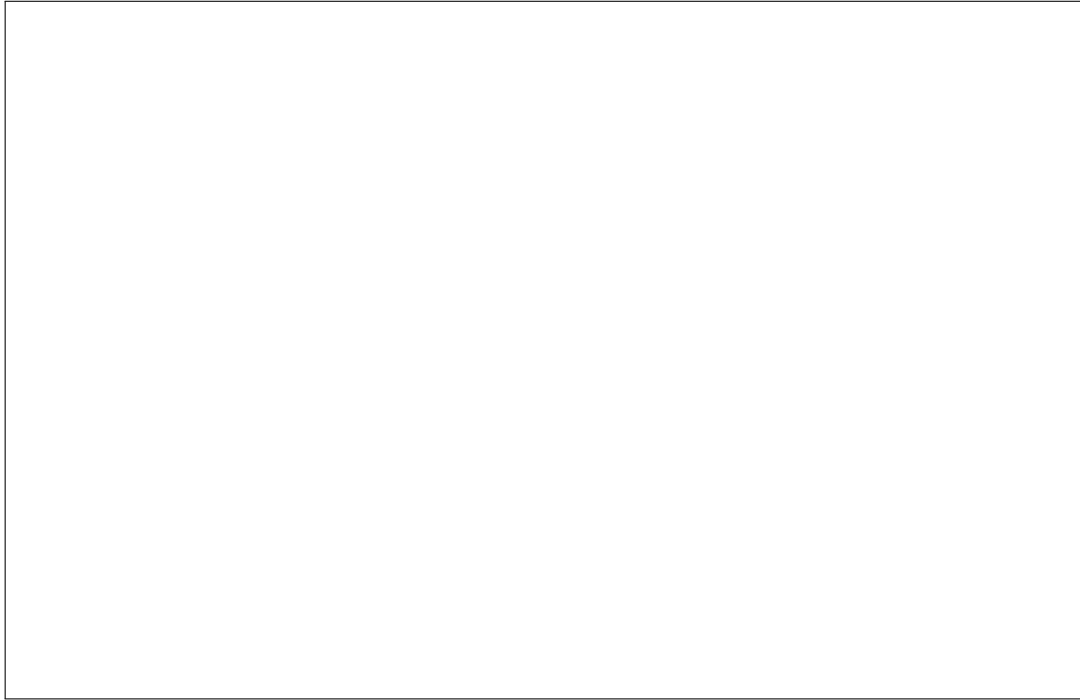
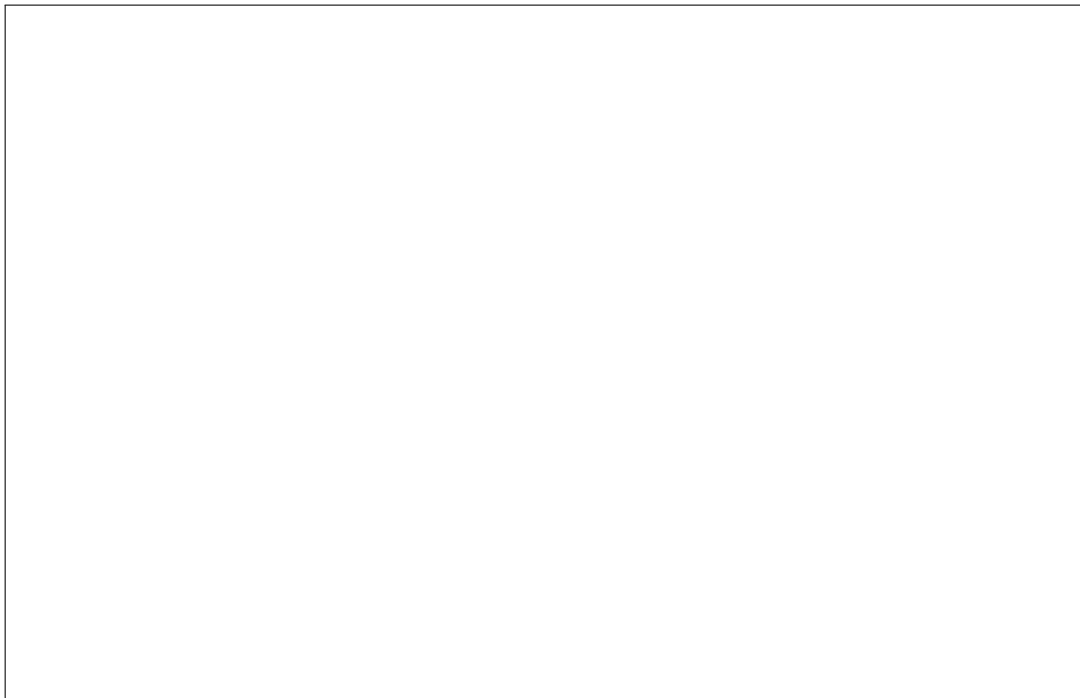
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Academic year: 2025/2026

Assessments and signature of the supervisors

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Abstract

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List of Abbreviations

AI	Artificial Intelligence
AI4DM	AI for Decision Making Lab, Concordia University
API	Application Programming Interface
BM25	Best Match 25, a probabilistic lexical ranking function
ENSI	National School of Computer Science (<i>École Nationale des Sciences de l'Informatique</i>)
EV	Engineering, Computer Science and Visual Arts Complex, Concordia University
FDR	False Discovery Rate
LLM	Large Language Model
MIAE	Mechanical, Industrial and Aerospace Engineering, Concordia University
NLI	Natural Language Inference
PAC	Probably Approximately Correct
PPI	Prediction-Powered Inference
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RAG	Retrieval-Augmented Generation
UQ	Uncertainty Quantification

General Introduction

Chapter 1

General Presentation and Context of the Internship

Introduction

The purpose of this chapter is to situate the project in its general framework, and to set out the problem it addresses, the objectives it pursued, and the manner in which the work was conducted.

The work reported here was carried out as a research internship at the AI for Decision Making Lab of Concordia University, under the supervision of Dr. Yassine Yaakoubi, and constitutes the research component of the final year of engineering studies at the Ecole Polytechnique de Tunisie.()

We first present the host organisation, its structure and its areas of research. We then state the problem the work addresses, in the terms in which it arises for a deployed system, and situate it in the systems already deployed and the regulations that govern them, before setting out the objectives pursued and the deliverables produced. The chapter closes with the methodology by which the work was conducted.

1.1 Host organisation

The internship was hosted by the **AI for Decision Making Lab (AI4DM)**, a research group attached to the Gina Cody School of Engineering and Computer Science at Concordia University, in Montreal, Canada. Table 1.1 summarises the host organisation; the subsections below present the host university, the faculty to which the laboratory belongs, and finally the research group in which the work was carried out.

Table 1.1: Host organisation details

Institution	Concordia University
Faculty	Gina Cody School of Engineering and Computer Science
Department	Mechanical, Industrial and Aerospace Engineering (MIAE)
Laboratory	AI for Decision Making Lab (AI4DM)
Supervisor	Dr. Yassine Yaakoubi
Location	Montreal, Quebec, Canada
Research axes	Optimisation, learning and simulation for large-scale decision making under uncertainty; large language models and retrieval-augmented generation; uncertainty quantification and reliability of AI systems

1.1.1 Concordia University

Concordia University is a public comprehensive research university located in Montreal, Quebec. It was founded in 1974 by the merger of two older institutions[N1]: Loyola College, established in 1896, and Sir George Williams University, which grew out of the evening classes offered by the Montreal YMCA from 1851[N2]. Its name comes from the motto of the city of Montreal, *Concordia salus*, that is to say “well-being through harmony”.

With 43,922 students enrolled in credit courses in 2024–2025 over its two campuses, Sir George Williams downtown and Loyola in the west of the city, Concordia is today one of the largest universities in Canada and one of the leading Canadian institutions in engineering and applied sciences[N3].

1.1.2 Gina Cody School of Engineering and Computer Science

The Gina Cody School of Engineering and Computer Science is the faculty of Concordia University dedicated to engineering and computer science. In 2018, it became the first engineering faculty in Canada to be named after a woman, following a donation of fifteen million dollars from Gina Parvaneh Cody[N4], who had been in 1989 the first woman to obtain a doctorate in building engineering at Concordia, and who has been the chancellor of the university since January 2025[N5].

The school gathers 11,315 students and 262 faculty members around 43 undergraduate and graduate programmes and 20 interdisciplinary research centres[N6]. It is organised into seven departments, among which the Department of Mechanical, Industrial and Aerospace Engineering (MIAE), to which the host laboratory belongs. Its teaching and research activities are housed principally in the Engineering, Computer Science and Visual Arts Complex shown in Figure 1.1.



Figure 1.1: The EV Complex, home of the Gina Cody School (photograph: Tasitch, CC BY 2.0)

1.1.3 Research group, supervision, and research axes

The work presented in this report was conducted within the AI for Decision Making Lab (AI4DM), the research group led by Dr. Yassine Yaakoubi, assistant professor in the MIAE department.

The research of the group is situated at the intersection of optimisation, learning and simulation for large-scale real-world decision making. It is organised around two main axes: machine-learning-augmented optimisation, which integrates learned models into classical optimisation methods, and end-to-end optimisation learning. To this end, the group combines mixed-integer and stochastic programming, decomposition methods, deep learning, reinforcement learning and graph neural networks, in order to design decision-support systems that are efficient, robust and scalable, with applications in logistics, transportation and sustainable supply chains.

A further strand of the group's activity extends this concern with robustness to systems built on large language models. It covers retrieval-augmented generation, where a model is grounded in evidence drawn from an external corpus, and the uncertainty quantification and reliability of such systems, that is, the question of how far their outputs can be trusted and how that trust can be measured. The work reported here belongs to this strand.

This scientific environment, centred on decision making under uncertainty and on the robustness of the resulting systems, is directly aligned with the subject of the internship, which aims at providing reliability guarantees for multi-agent question-answering systems.

1.2 Problem statement

1.3 Objectives and expected deliverables

1.4 Working methodology

Conclusion

Chapter 2

Background and literature review

Introduction

This chapter sets out the technical background and the literature on which the rest of the report builds. Its aim is to introduce the notions of retrieval-augmented generation used throughout, and to follow the question of the reliability of these systems up to the point where the problem treated during the internship appears.

We first present the principles and architecture of retrieval-augmented generation and the settings in which it is deployed. We then examine the reliability challenges of these systems and review the responses proposed to them, from uncertainty quantification to verification. We finally turn to the statistical guarantees that can be attached to their outputs, before synthesising this literature and setting out the question that Chapter 3 takes up.

Conclusion

Chapter 3

A Systematic Survey of the Trust Layer of RAG

Introduction

Conclusion

Chapter 4

chapter 4

Introduction

Conclusion

Chapter 5

chapter 5

Introduction

Conclusion

Conclusion and outlook

This internship project aimed to design

Netography

- [N1] Concordia University, “The birth of Concordia: Loyola College and Sir George Williams University merged on 24 August 1974.” <https://www.concordia.ca/cunews/main/stories/2015/08/24/archives-loyola-and-sir-george-williams-merge.html>. Accessed: 2026-08-13.
- [N2] Concordia University, “Sir George Williams University history.” <https://www.concordia.ca/offices/archives/stories/sgw.html>. Accessed: 2026-08-13.
- [N3] Concordia University, “Concordia fast facts: student enrolment 2024–25.” <https://www.concordia.ca/about/fast-facts.html>. Accessed: 2026-08-13.
- [N4] Concordia University, “First engineering faculty in Canada named after a woman: business leader Gina Cody makes historic \$15 million donation.” <https://www.concordia.ca/news/stories/2018/09/24/gina-cody-makes-historic-eponymous-15m-donation-to-have-canadas-first-engineering-faculty-named-after-a-woman.html>. Published 24 September 2018. Accessed: 2026-08-13.
- [N5] Concordia University, “Gina Cody — notable Concordians.” <https://www.concordia.ca/profiles/offices/advancement/notable-concordians/gina-cody.html>. Accessed: 2026-08-13.
- [N6] Concordia University, “About the school — gina cody school of engineering and computer science.” <https://www.concordia.ca/ginacody/about.html>. Accessed: 2026-08-13.